

Genome-Wide Association Exercise

Association Analysis Controlling for Population Substructure

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1. Population Stratification and Association Testing

The dataset you cleaned earlier in the week actually comes from HapMap Phase III data. It consisted before cleaning of CEU founders (Caucasians from Utah), MEX founders (Mexicans from Los Angeles) and TSI (Tuscans from Italy) with some additional data tweaking for you to have things to do in the exercise. The CEU pedigree identifiers begin with all numbers e.g. 1347, the MEX pedigree identifiers all start with M e.g. M017 and the TSI pedigree identifiers all start with NA e.g. NA0217. Before we start testing for association, we want to know if there is population stratification that we will need to control. If not, we risk observing an association, which is due to a difference in genotype frequencies in cases and controls, because of population substructure and not because of linkage disequilibrium between test SNP and the functional variant. We are going to use multidimensional scaling (MDS) within PLINK software to generate 10 components for admixture in the study. Please note that PLINK uses MDS instead of principal components, but they are almost identical. **Disclaimer: You should never collect data from European-Americans, Mexican-Americans and Italians and attempt to analyze them together even if you control for population stratification.**

Note: With a real dataset, you will likely have genome-level coverage and you should first prune your SNPs to obtain a subset in approximate linkage equilibrium prior to performing MDS on the data. For more information on how to do this in PLINK look at

<http://pngu.mgh.harvard.edu/~purcell/plink/summary.shtml> under “Linkage disequilibrium based SNP pruning”

```
plink --file GWAS_clean4 --genome --cluster --mds-plot 10
```

This command outputs the file **plink.mds** that contains the subject IDs and values for the 10 components we just generated. There is another file in your folder called **mds_components.txt**. This file is identical to your **plink.mds** file with the exception of a Group column that codes the CEU individuals as 1, the MEX individuals as 2 and the TSI individuals as 3. This is done so when we plot the components in R you can see what group the points belong to and judge for yourself how well you are clustering the data with the MDS components. The following commands will generate a jpeg image file containing the mds plot (filename=mds.jpeg) in your current working directory. Open R and put in the following code:

```
mydata = read.table("mds_components.txt", header=T)
```

```
mydata$pch[mydata$Group==1 ] <-15  
mydata$pch[mydata$Group==2 ] <-16  
mydata$pch[mydata$Group==3 ] <-2
```

```
jpeg("mds.jpeg", height=1000, width=1000)  
plot(mydata$C1, mydata$C2 ,pch=mydata$pch)  
dev.off()
```

Now going back to PLINK, we are going to use as covariates in a logistic regression model the components that were generated by MDS and the individual SNPs in the study to calculate the genomic inflation factor (λ). When you analyze SNPs for association with your trait you generate a

test statistic for each SNP and λ is the ratio of the median of these test statistics to the expected median of the test statistics. If there were no population stratification λ would be 1. So we will use this measure to decide how many components from our MDS analysis we will add in the logistic regression model for each SNP. First, let find out what λ is without adjusting:

```
plink --file GWAS_clean4 --pheno pheno.txt --pheno-name Aff --logistic --adjust -
-out unadj
```

And then find out what λ is when we adjust for the first component:

```
plink --file GWAS_clean4 --pheno pheno.txt --pheno-name Aff --covar plink.mds --
covar-name C1 --logistic --adjust --out C1
```

And the first and second components:

```
plink --file GWAS_clean4 --pheno pheno.txt --pheno-name Aff --covar plink.mds --
covar-name C1-C2 --logistic --adjust --out C1-C2
```

and so forth for all 10 components in the plink.mds file completing the table:

Table 1

	Un- adjusted	C1- C1	C1- C2	C1- C3	C1- C4	C1- C5	C1- C6	C1- C7	C1- C8	C1- C9	C1- C10
λ	1.12	1.08	1.02	1.02	1.03	1.03	1.02	1.03	1.05	1.06	1.07
Mean chisquare											

You can see that above λ reaches the optimal value of 1.0 after a certain number of components are included. This is the number of components you should be adjusting for.

Go to the **assoc.logistic file that corresponds to that number of components** and make a note of how you named the **.assoc.logistic** file for it and when you did not adjust for any components. Then go back to the R program to load the results and create a jpeg image file containing QQ plots for the adjusted and unadjusted results (using a modified script from <http://www.broad.mit.edu/node/555>) as follows:

```
broadqq <-function(pvals, title)
{
  observed <- sort(pvals)
  lobs <- -(log10(observed))

  expected <- c(1:length(observed))
  lexp <- -(log10(expected / (length(expected)+1)))

  plot(c(0,7), c(0,7), col="red", lwd=3, type="l", xlab="Expected (-logP)", ylab="Observed (-logP)",
xlim=c(0,max(lobs)), ylim=c(0,max(lobs)), las=1, xaxs="i", yaxs="i", bty="l", main = title)
  points(lexp, lobs, pch=23, cex=.4, bg="black") }

jpeg("qqplot_compare.jpeg", height=1000, width=1000)
par(mfrow=c(2,1))
aff_unadj<-read.table("unadj.assoc.logistic", header=TRUE)
aff_unadj.add.p<-aff_unadj[aff_unadj$TEST==c("ADD"),]$P
broadqq(aff_unadj.add.p,"Some Trait Unadjusted")
aff_C1C2<-read.table("C1-C2.assoc.logistic", header=TRUE)
```

```
aff_C1C2.add.p<-aff_C1C2[aff_C1C2$TEST==c("ADD"),]$P
broadqq(aff_C1C2.add.p, "Some Trait Adjusted")
dev.off()
```

Now look for SNPs with genome-wide significance with the following in R:

```
gws_unadj = aff_unadj[which(aff_unadj$P < 0.0000001),]
gws_unadj
gws_adjusted = aff_C1C2[which(aff_C1C2$P < 0.0000001),]
gws_adjusted
```

Question 1:

Did this study have a finding with genome-wide significance after adjusting for population substructure? Did you notice any difference in the p-values before and after adjustment for substructure? Also, complete the tables below.

Yes, there were 2 SNPs that were significant. They were still significant before adjustment but the p-values changed.

Using the QQ plots, we can see that the outliers don't change much, but many of the middle points are more in line.

This reflects the data in our tables.

Table 2. SNPS with genome-wide significance unadjusted for substructure:

CHR	SNP	BP	A1	TEST	NMISS	OR	STAT	P
4	rs10008252	179853616	G	ADD	244	0.16650	-6.639	3.155e-11
8	rs4571722	60326734	T	ADD	242	0.04126	-7.436	1.037e-13

Table 3. SNPs with genome-wide significance adjusted for components 1 and 2:

CHR	SNP	BP	A1	TEST	NMISS	OR	STAT	P
4	rs10008252	179853616	G	ADD	244	0.13250	-6.688	2.255e-11
8	rs4571722	60326734	T	ADD	242	0.04363	-7.241	4.459e-13

Question 2: Why would you not want to include in your analysis individuals from different ethnic backgrounds even if you control for population substructure?

Controlling for population substructure is not a perfect method. There are still other factors that could influence the results. Even if the same genes are present, different subpopulations may have different functional variants or allele frequencies that could be a confounding factor in our analysis.

A common question when you have a finding with genome-wide significance in a GWAS is “Is the SNP in a known gene?” One way to look this information up is to query the SeattleSNPs Genome Variation Server. Go to the website:

<http://gvsbatch.gs.washington.edu/GVSBatch/>

Now open up the file, seattle_batch.txt and type the rs numbers of the SNPs with genome-wide significance. Then upload this file to the website above and give it your correct email address. In a few minutes you will receive an email with the results of your query.

Question 3. Are any of your SNPs with genome-wide significance in known genes?

No, they are intergenic.

1. Population Stratification and Association Testing

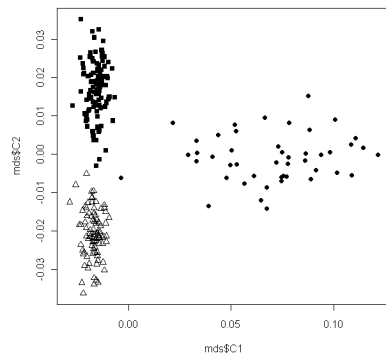


Table 1

	Un-adjusted	C1	C1-C2	C1-C3	C1-C4	C1-C5	C1-C6	C1-C7	C1-C8	C1-C9	C1-C10
Λ Mean chisqare	1.11	1.08	1.02	1.02	1.02	1.02	1.03	1.03	1.05	1.06	1.06
	1.08	1.03	1.00	1.01	1.01	1.02	1.02	1.03	1.03	1.04	1.04

Answer to Question 1:

Question 1:

Did this study have a finding with genome-wide significance after adjusting for population substructure? Did you notice any difference in the p-values before and after adjustment for substructure?

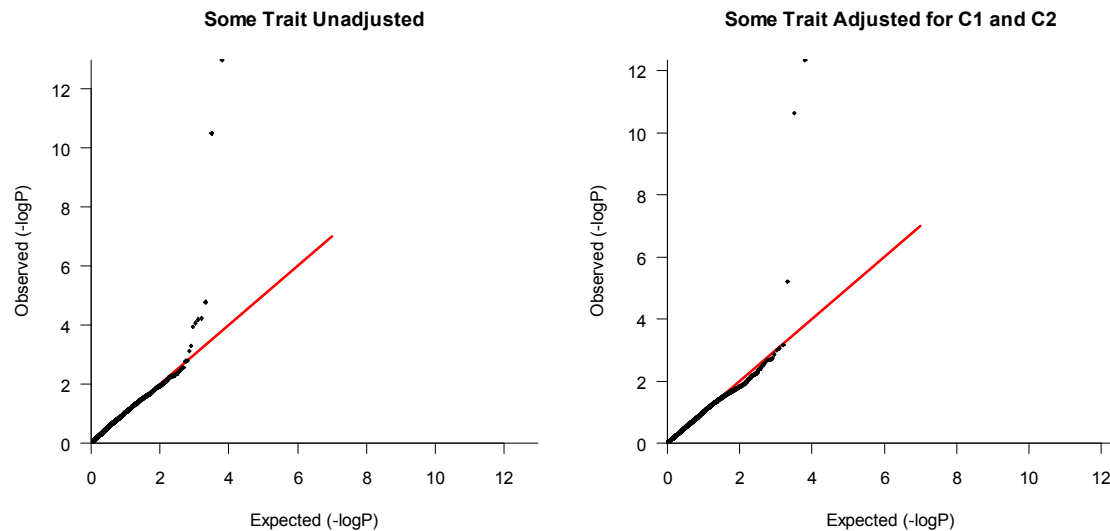
Yes, see tables below. If you compare the QQ plots below you can see that for this dataset the most significant SNPs were changed minimally when we adjusted for substructure but some of the moderately significant SNPs became less significant after adjustment.

Table 2. SNPS with genome-wide significance unadjusted for substructure:

CHR	SNP	BP	A1	TEST	NMISS	OR	STAT	P
8	rs4571722	60326734	T	ADD	242	0.04126	-7.436	1.04E-13
4	rs10008252	179853616	G	ADD	244	0.1665	-6.639	3.16E-11

Table 3. SNPs with genome-wide significance adjusted for components 1 and 2:

CHR	SNP	BP	A1	TEST	NMISS	OR	STAT	P
8	rs4571722	60326734	T	ADD	242	0.04362	-7.241	4.46E-13
4	rs10008252	179853616	G	ADD	244	0.1325	-6.688	2.26E-11



Question 2: Why would you not want to include in your analysis individuals from different ethnic backgrounds even if you control for population substructure?

Firstly, you may not be able to adequately control for population substructure. Secondly, even if within the different populations the same genes are involved, the same functional variants may not be involved therefore the same tagSNPs will not pick-up associations with different functional variants within the different populations. In this situation power to detect an association can be greatly reduced. Even if within the different populations the same functional variant(s) are involved in disease etiology the tagSNPs in the different populations can have different allele frequencies, therefore the functional variant will not be tagged equally well in all populations and the power can be reduced.

Question 3. Are any of your SNPs with genome-wide significance in known genes?

No, both rs457122 and rs10008252 are intergenic.